

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate specializing in scalable likelihood inference, predictive modeling, and statistical diagnostics for stochastically dependent data. Develops rigorous hypothesis-driven research workflows and reproducible Python/PyTorch tools for large datasets, with C++/pybind11 integration for performance-critical computation.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected May 2027 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Research Systems: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), C++/pybind11 integration, R, SQL, Linux, Git, AWS EC2, HPC/SLURM; CPU/GPU computing and restartable research pipelines

Quantitative Methods: Hypothesis testing, predictive modeling, regression and machine learning, backtesting, probability and stochastic processes, Monte Carlo simulation, MLE/GLS, numerical optimization, time-series and spatiotemporal modeling, spectral and autocorrelation analysis, model validation, and performance evaluation

QUANTITATIVE INDUSTRY EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review

Summer 2026

- Developed a sequential risk-monitoring framework for binary indicators, using Brownian-motion approximation, dynamic-programming probability propagation, and simulation to quantify stopping-time uncertainty and early-decision tradeoffs.
- Specified and validated a Sequential Probability Ratio Test with explicit hypotheses and likelihood-ratio boundaries; evaluated boundary-crossing probabilities, stopping behavior, and Type I/II error.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pipeline across 2.46M+ records, including data preparation, feature processing, GLM benchmarking, model selection, validation, and performance diagnostics on AWS EC2; presented findings as pricing and risk-segmentation recommendations to business stakeholders.

QUANTITATIVE RESEARCH PROJECTS

Ph.D. Dissertation Research - Scalable Stochastic Modeling and Statistical Diagnostics

Sep 2024 - Present

- **Scalable Inference:** Developed an advection-aware Vecchia likelihood approximation for nonseparable space-time covariance structures, using ordered local conditional models to support likelihood-based parameter estimation and uncertainty quantification for up to 145,008 observations per day.
- **Stochastic Diagnostics:** Developed statistical diagnostic tools for dependent stochastic processes that localize where a fitted model fails across scale and frequency, distinguishing persistent dependence from high-frequency noise, missingness, acquisition artifacts, and covariance misspecification.
- **Research Tools:** Built reusable Python/PyTorch workflows for data-quality filtering, time-dependent alignment, optimization, simulation, and restartable CPU/GPU and HPC execution; integrated performance-critical C++ ordering routines through pybind11.

IMC Prosperity Algorithmic Trading Competition - Top 2.39% of Participants

Apr 2025

- Built and backtested an inventory-aware market-making strategy around empirically calibrated reference values; tracked position-limit saturation over a rolling 10-step window and triggered more aggressive inventory-reducing quotes to manage persistent exposure.

ADDITIONAL

Coursework: Machine Learning, Linear Algebra, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)